Types of Classification Error

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected confusion-matrix record.

Place IDs, name error types, audit equal-score matrices and limit a zero-error claim.

Keep one habit: Keep the paired records, axis convention, cell counts and remaining questions together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Accuracy as a Fraction

Human Classifier Simulation

Rows Columns and Data Types

W1 · Place IDs before writing counts

> Give both axis headers and the label order.

Your evidence record

- Worked reference: V1,V2 clear/clear; V3 clear/blocked; V4,V5 blocked/clear; V6,V7,V8 blocked/blocked. Each pair is reference then prediction. Rows reference, columns predicted, both ordered clear then blocked. Blocked is positive. Each ID contributes exactly once; the diagonal counts matches. Keep this eight-record case distinct from the separate six-book sound/repair case.

- Use all eight worked IDs. Draw a matrix with reference rows and **prediction** columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Explain the two error types

> Name the chosen positive class before using abbreviations.

Your evidence record

- Worked reference: V1,V2 clear/clear; V3 clear/blocked; V4,V5 blocked/clear; V6,V7,V8 blocked/blocked. Each pair is reference then prediction. Rows reference, columns predicted, both ordered clear then blocked. Blocked is positive. Each ID contributes exactly once; the diagonal counts matches. Keep this eight-record case distinct from the separate six-book sound/repair case.

- Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit totals and equal-score errors

> Compare the same reference cases and retain the full matrices.

Your evidence record

- Worked reference: V1,V2 clear/clear; V3 clear/blocked; V4,V5 blocked/clear; V6,V7,V8 blocked/blocked. Each pair is reference then prediction. Rows reference, columns predicted, both ordered clear then blocked. Blocked is positive. Each ID contributes exactly once; the diagonal counts matches. Keep this eight-record case distinct from the separate six-book sound/repair case.

- Give the worked row totals, column totals, total count and accuracy. Compare its errors with [[3,0],[3,2]] using the same axes and eight reference records.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit a zero-count claim

> Request the missing scope, headers and class definitions.

Your evidence record

- A report gives zero **false positive**s but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Count a fresh evaluation independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Count a fresh evaluation independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Separate recommending an unsuitable book from failing to surface a suitable one.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Explain the two error types for Confusion matrices. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Positive means sports. A sports book predicted other and a non-sports book predicted sports occur. Name each error and its consequence for the sports shelf.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 13 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

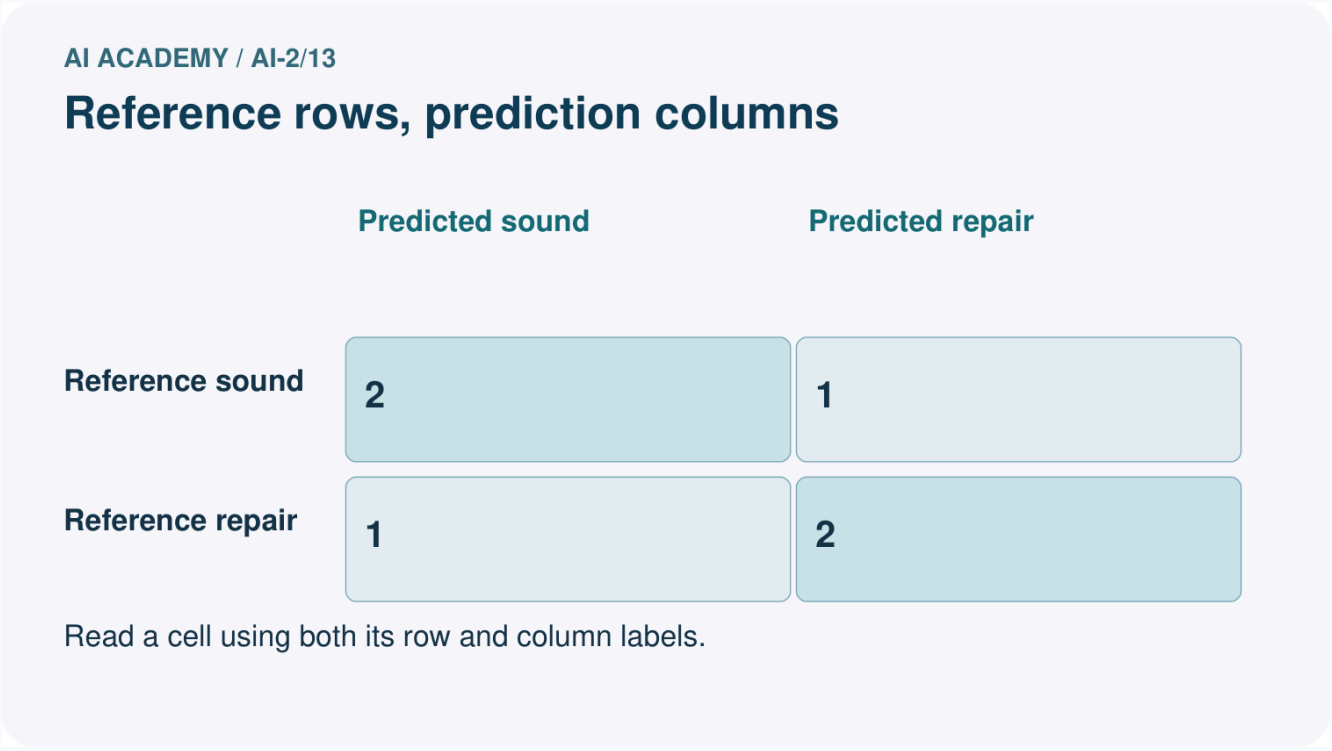
> Keep the explanation and the support you used.

Use this when helpful

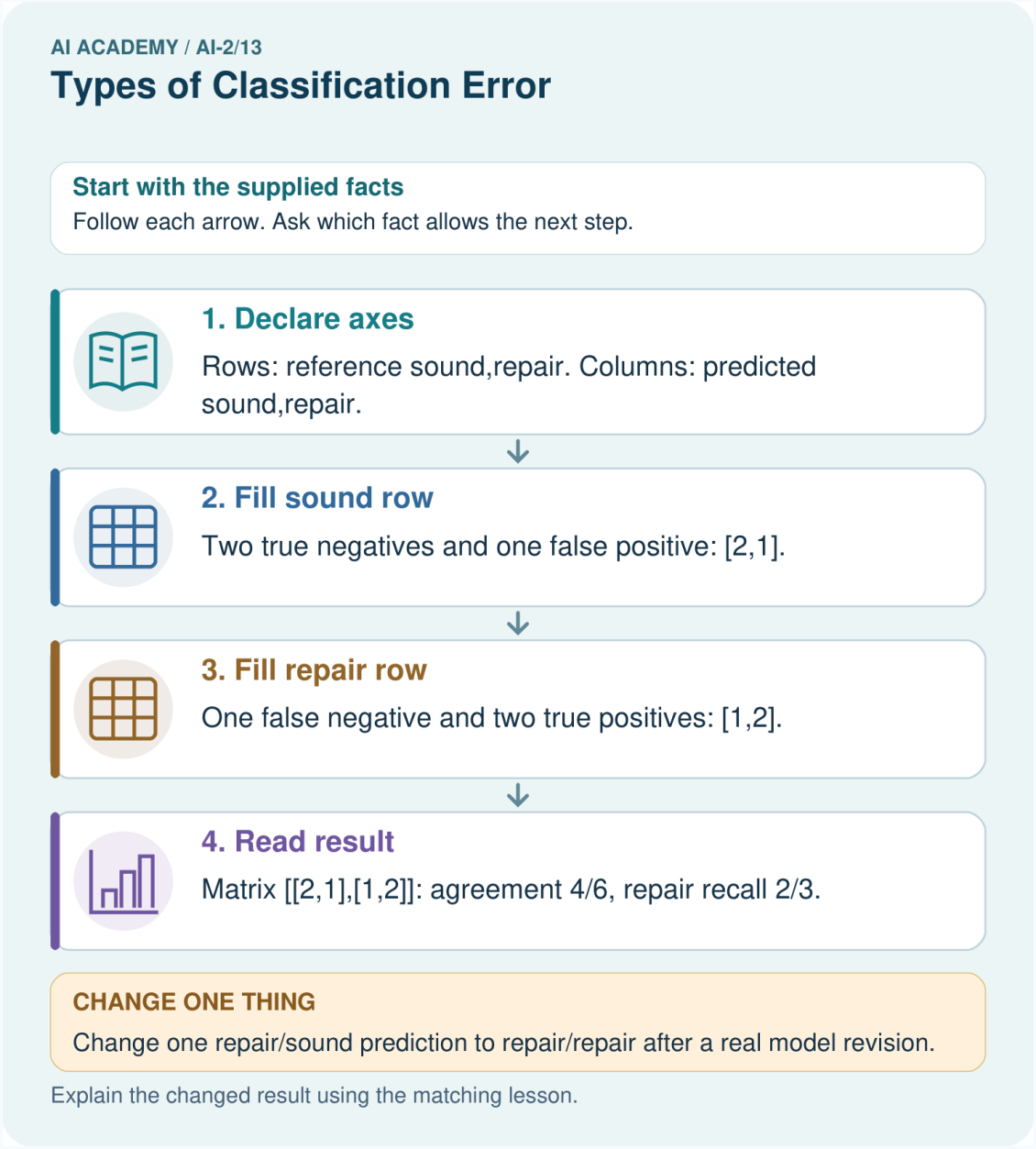
A fictional book-condition checker predicts repair or sound.

Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Which error causes a sound book to be sent for unnecessary repair review?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional book-condition checker predicts repair or sound.
- Which labelled step matches this detail? Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.
- 1. Declare axes
- 2. Read result
- 3. Fill repair row
- 4. Fill sound row
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Read result Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Each case enters one cell according to its reference and predicted label.

The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Changed-case question

- Changed-case practice: Change one repair/sound **prediction** to repair/repair after a real model revision. Which conclusion is supported by the supplied facts?
- 1. Repair row becomes [0,3]; agreement becomes 5/6.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Repair row becomes [0,3]; agreement becomes 5/6. One **false negative** moves to **true positive**; the sound row stays [2,1].

Repair and save your work

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Types of Classification Error: the case

AI ACADEMY / AI-2/13 · 01 · LOOK AT THE FACTS

Types of Classification Error: the case

The situation

A fictional book-condition checker predicts repair or sound.

Supplied evidence

Use reference rows and prediction columns in order sound,repair. Repair is positive.
Six reference/prediction pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

By the end, I can construct a confusion matrix with declared axes and name both error types.

Follow the labelled picture with the matching explanation. Open the vector diagram

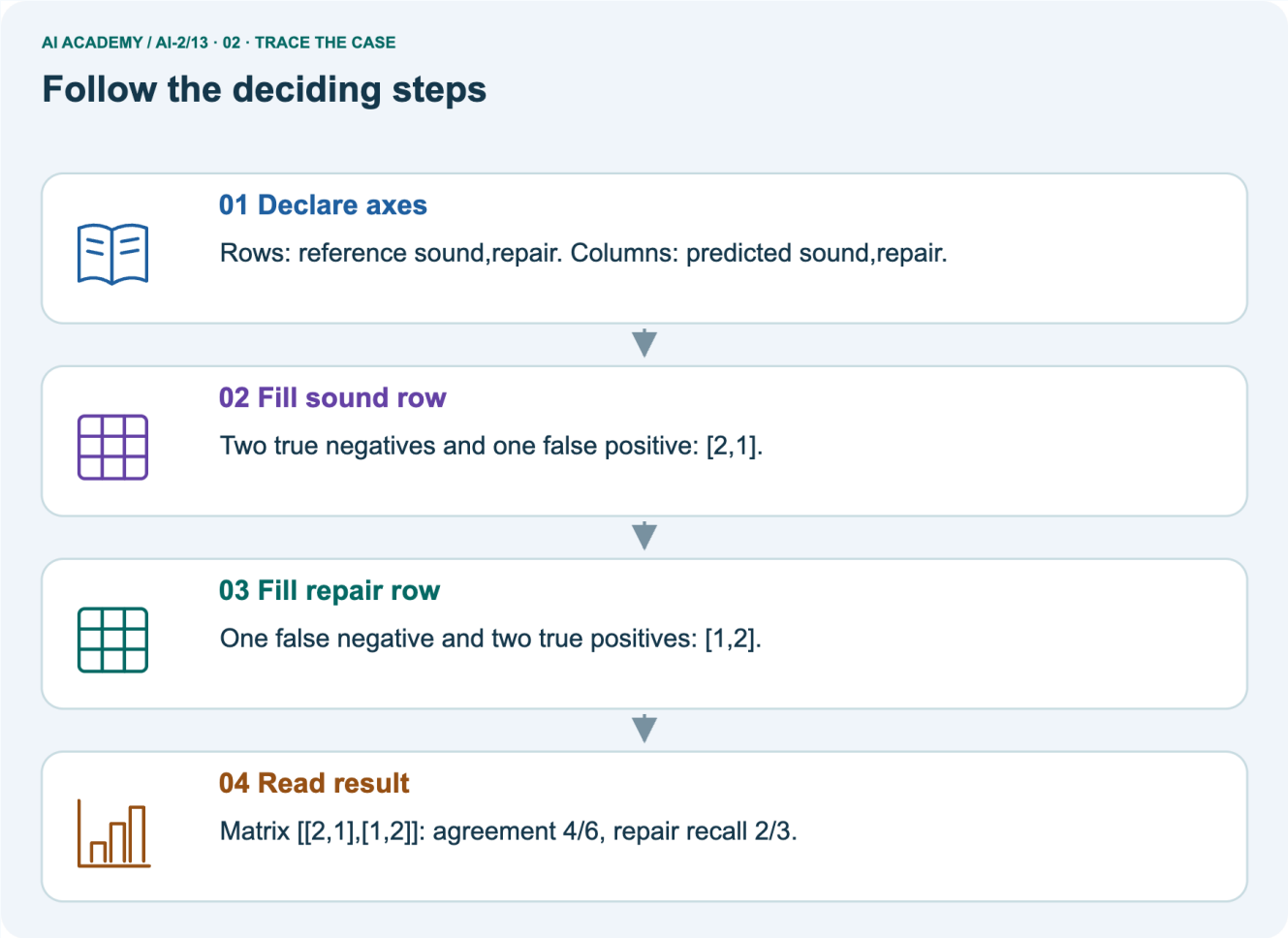
A fictional book-condition checker predicts repair or sound. Supplied facts: Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

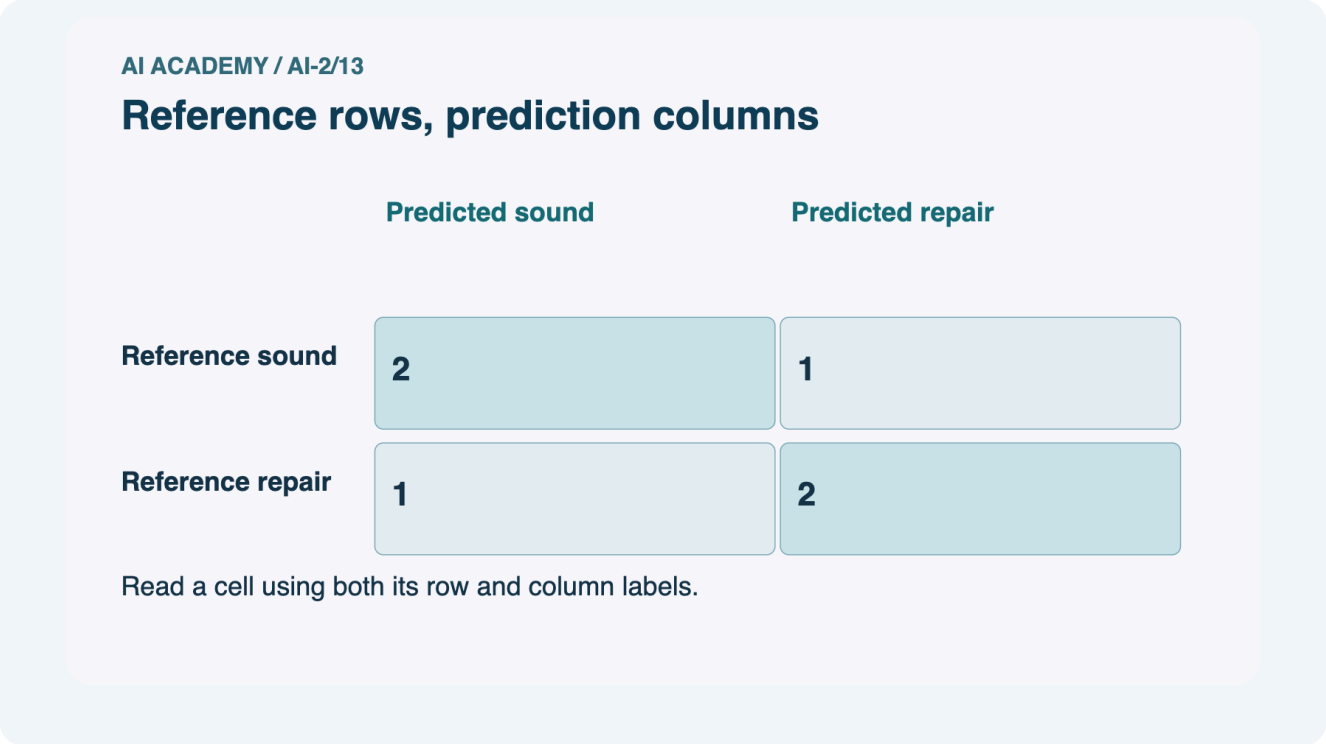
Declare axes: Rows: reference sound,repair.

Columns: predicted sound,repair. → Fill sound row: Two **true negative**s and one **false positive**: [2,1]. → Fill repair row: One **false negative** and two **true positive**s: [1,2]. → Read result: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Reference rows, prediction columns



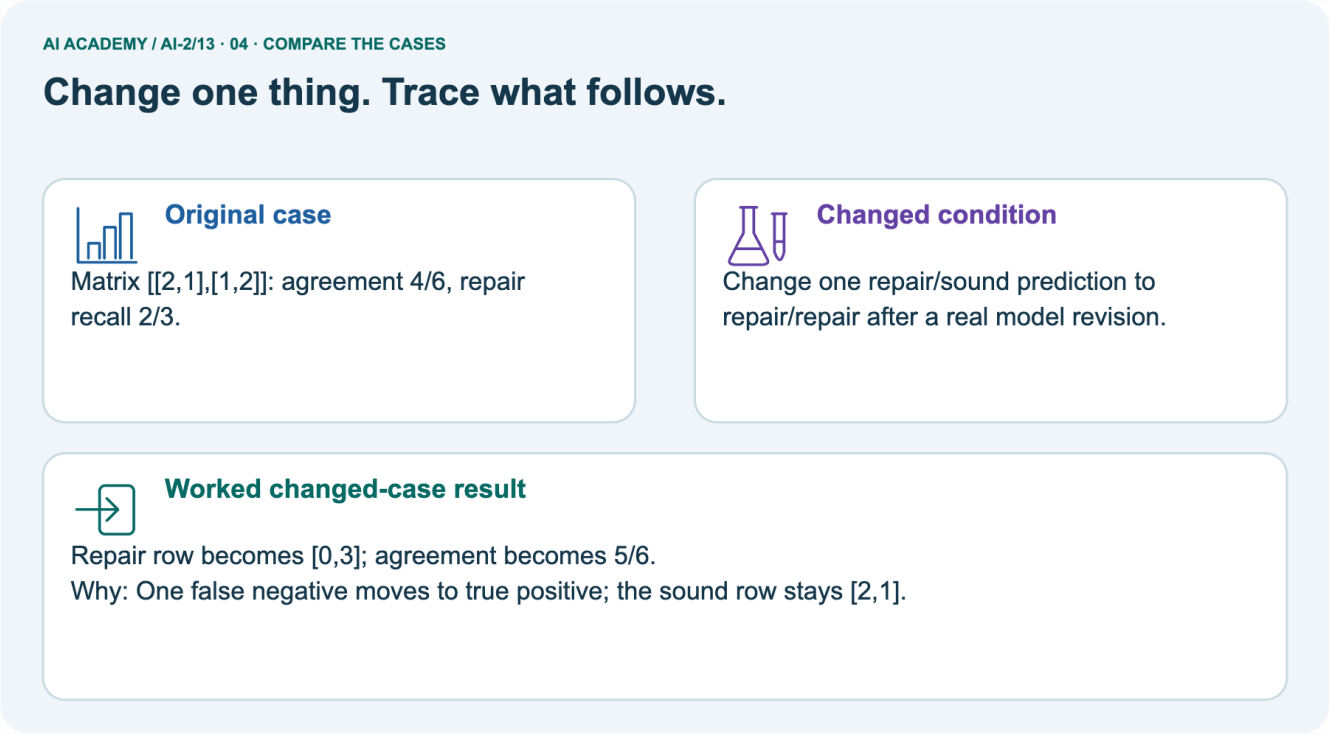
Follow the labelled picture with the matching explanation. Open the vector diagram

Reference rows, **prediction** columns. Reference sound: Predicted sound = 2, Predicted repair = 1. Reference repair: Predicted sound = 1, Predicted repair = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

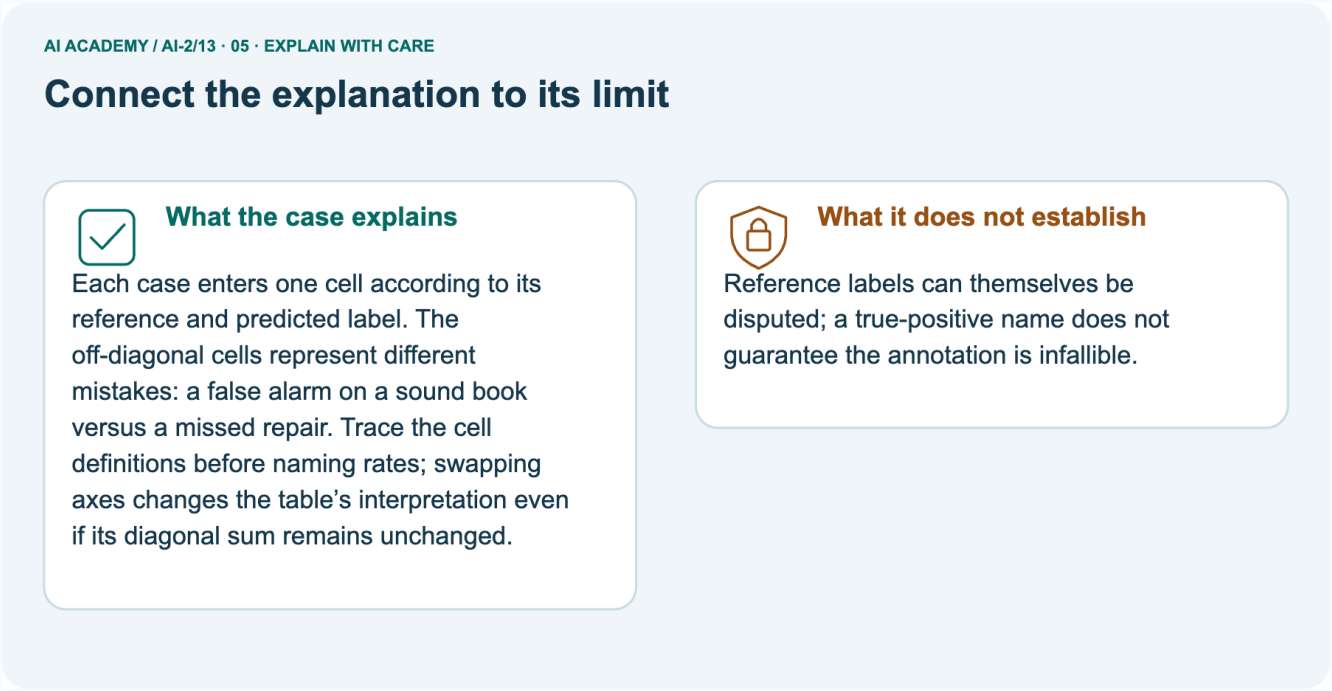
Original case: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Changed condition: Change one repair/sound **prediction** to repair/repair after a real model revision.

Expected result: Repair row becomes [0,3]; agreement becomes 5/6. Reason: One **false negative** moves to **true positive**; the sound row stays [2,1].

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

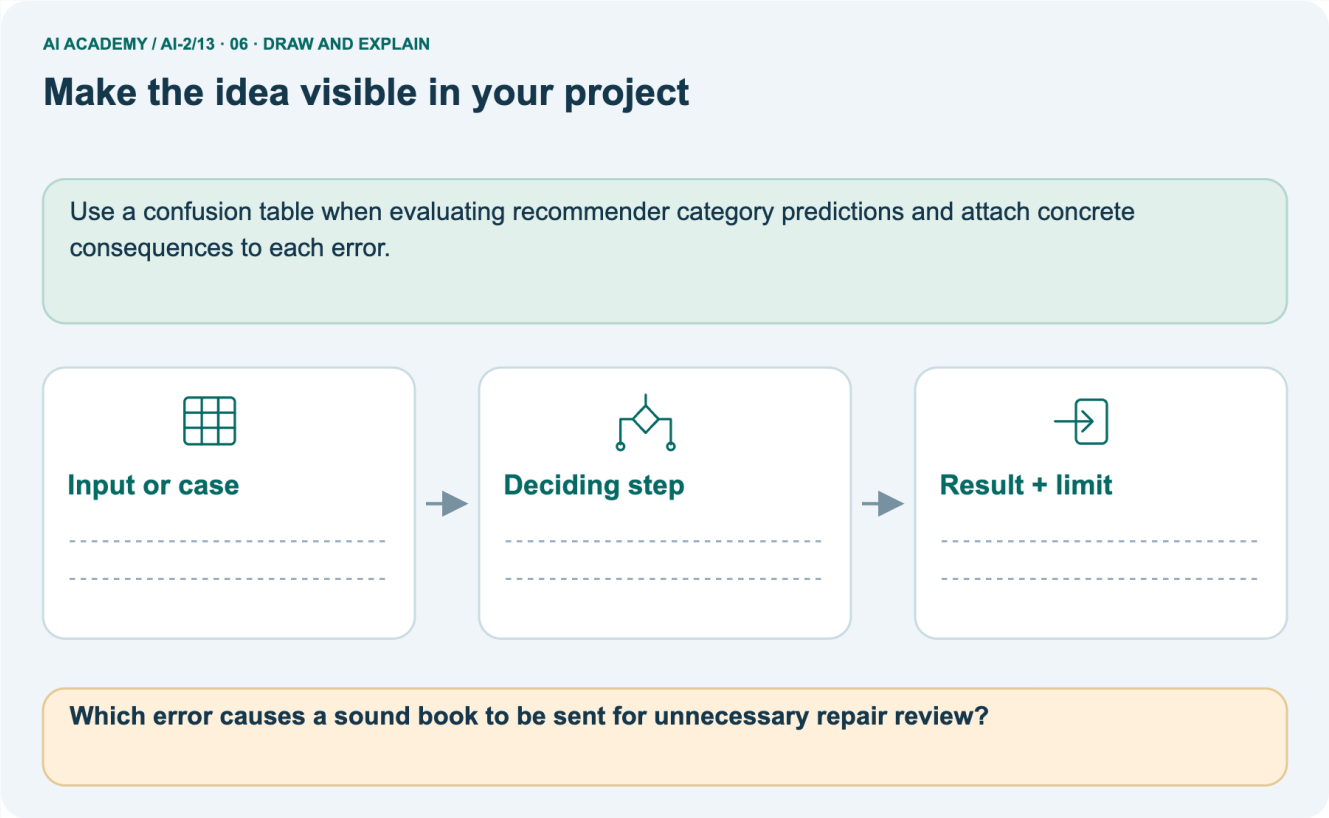
Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Limit: Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error. Draw your **input**, deciding step, result and limit.

Supported question: Which error causes a sound book to be sent for unnecessary repair review?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Types of Classification Error: the case

02. Follow the deciding steps

03. Reference rows, prediction columns

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.